# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular Examination October 2025

**Third Semester**

**Discipline Specific Core Course (DSC) - MG3DCSFSQ200 - FOOD PRESERVATION AND ADDITIVES**

(2024 Admission Onwards)

Duration: 2 Hours Maximum Marks: 50

***Remember(R), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

**Part A**

**MCQ**

Answer all questions

*Each question carries 1 mark*

1. Describe one limitation of sterilization in food preservation.
a) Loss of some nutrients and sensory qualities

2. Identify the typical pressure range used in HPP for food preservation.
c) 300–600 MPa

3. The main principle of concentration is
b) Reduction of water activity and volume

4. The compression valve in refrigeration cycle functions to
d) Compress refrigerant vapor

5. The material commonly used for electrodes in Ohmic heating system is
c) Stainless steel

6. Select major limitation of canning is
b) High cost and nutrient loss

7. Select the component of food that absorbs microwave energy
c) Water molecules

8. Enzyme inactivation in foods by heat is important to prevent
b) Ripening and spoilage

9. Name the permitted additive would you apply to soups to enhance umami flavor?
b) Monosodium glutamate

10. To balance acidity in fruit juices, which additive would be applied?
b) Acidulants

( 10x1mark= 10)

**Part B**

Short Answers

Answer Any 4

*Each question carries 5 marks*

11. Classify the types of pasteurization based on temperature and time.

- Pasteurization is the process of heating foods, mainly liquids, to destroy pathogenic microorganisms without affecting the nutritional and sensory qualities.
- Based on temperature and time, pasteurization is classified as:
    - Low Temperature Long Time (LTLT): Heating at 63°C for 30 minutes – commonly used for milk and liquid foods.
    - High Temperature Short Time (HTST): Heating at 72°C for 15 seconds – rapid process ensuring microbial destruction while maintaining flavor.
    - Ultra High Temperature (UHT): Heating at 135–150°C for 1–2 seconds – sterilizes product for long shelf life without refrigeration.
- The choice depends on product type, microbial load, and desired shelf life.

12. Describe the different types of membranes used in food applications.

- Membrane separation processes are used for concentration, purification, and fractionation of food materials.
- Major types include:
    - Microfiltration (MF): Pore size 0.1–10 µm; removes suspended solids and bacteria.
    - Ultrafiltration (UF): Pore size 0.001–0.1 µm; separates proteins and macromolecules.
    - Nanofiltration (NF): Pore size 0.0001–0.001 µm; removes salts and small organics.
    - Reverse Osmosis (RO): Smallest pores; removes water from solutions for concentration.
- Applications include whey protein concentration, juice clarification, and water purification.

13. Discuss the effect of food additives on human health.

- Food additives enhance flavor, appearance, texture, and shelf life of foods.
- Positive effects:
    - Preservatives prevent spoilage and extend shelf life.
    - Antioxidants reduce oxidation and maintain nutritional quality.
    - Fortifying agents improve nutritional value.
- Negative effects:
    - Excess artificial additives may cause allergies or hyperactivity in children.
    - Some synthetic colorants and preservatives may have carcinogenic potential if misused.
- Regulatory agencies (FSSAI, FDA) set safe limits for additive use to protect consumers.

14. Interpret the role of temperature in food preservation.

- Temperature is a factor controlling microbial growth, enzyme activity, and chemical reactions in food.
- Low temperature preservation:
    - Refrigeration (0–5°C) slows microbial activity and enzyme reactions.
    - Freezing (-18°C or lower) halts microbial growth by converting water into ice.
- High temperature preservation:
    - Pasteurization and sterilization destroy pathogenic and spoilage microorganisms.
    - Blanching inactivates enzymes before freezing or canning.
- Proper temperature control ensures food safety and retains sensory and nutritional quality.

15. A food label shows E330. Apply your knowledge to identify the additive and explain its role in foods.

- E300-E400 comes under Antioxidant
- E330 is the code for Citric Acid, a natural organic acid widely used in food processing.
- It is obtained naturally from citrus fruits or produced by fermentation of sugars using Aspergillusniger.

Functions:

- Acts as an acidulant to control pH and impart a tart flavor.
- Serves as an antioxidant by chelating metal ions and preventing oxidation.
- Enhances the effectiveness of preservatives like sodium benzoate.
- Used in beverages, jams, jellies, and fruit juices to maintain flavor and color.

Safe within prescribed limits; excessive intake may cause dental erosion.

Role of E330 (Citric Acid) in Foods:

- Acidulant::Maintains the desired acidity (pH) in foods and beverages.Provides a refreshing sour or tangy taste that enhances flavor.
- Preservative:Inhibits the growth of spoilage microorganisms by lowering the pH.,Extends the shelf life of fruit juices, jams, and sauces.
- Antioxidant:Acts as a chelating agent that binds metal ions (like iron and copper) which catalyze oxidation.
- Flavor Enhancer and Stabilizer:Balances sweetness and sourness in beverages and confectionery.Helps retain color, texture, and aroma during processing and storage.
- Synergistic Effect with Other Additives:Boosts the effectiveness of other preservatives such as sodium benzoate and sorbates.

16. Explain physical methods of sterilization.

- Sterilization is the process of destroying all microorganisms, including spores, to ensure food safety.
- Physical methods include:
    - Heat sterilization: Using moist heat (autoclaving) or dry heat (hot air oven) to destroy microbes.
    - Filtration: Physically removing microorganisms from liquids using membrane filters.
    - Radiation: Gamma rays or UV rays used to sterilize packaged or surface foods.

    - High-pressure processing (HPP): Inactivates microbes without heat, preserving nutrients.
- These methods ensure microbial safety while maintaining product quality depending on application.
  ( 4x 5 marks=20)

**Part C**

**Essay**

***Answer any 2***

***10 Marks Each***

17. Elaborate the principle, process, equipment and application of food irradiation in preservation.

- Food irradiation is a process of exposing food to controlled ionizing radiation such as gamma rays, X-rays, or electron beams to destroy microorganisms, insects, and parasites.
- Principle: Radiation breaks the DNA of microorganisms and insects, preventing reproduction and growth, thus extending shelf life without raising temperature.
- Types of radiation: Gamma rays (Cobalt-60, Cesium-137), X-rays, and Electron beams from linear accelerators.
- Process: Packed food is passed through an irradiation chamber and exposed to a controlled radiation dose (measured in kGy).
- Equipment: Gamma irradiator, linear accelerator, and conveyor systems are used for uniform exposure.
- Radiation dose categories: Low dose (up to 1 kGy) for disinfestation, Medium (1–10 kGy) for pathogen reduction, High (>10 kGy) for sterilization.
- Applications: Control insects in grains, delay ripening, reduce pathogens in meat, sterilize ready-to-eat foods.
- Advantages: Extends shelf life, maintains freshness, safe and residue-free process.
- Limitations: High equipment cost, slight vitamin loss, and consumer misconceptions about radiation safety.
- Safety: Approved by WHO, FAO, and FSSAI when used within prescribed limits.

18. Explain the importance of antioxidants and their classification in food processing.

- Antioxidants are compounds that delay or prevent oxidation of fats, oils, and vitamins in foods.
- They prevent rancidity, maintain color and flavor, protect nutrients, and extend shelf life.
- Mechanism: Antioxidants donate hydrogen or electrons to free radicals, terminating oxidation chain reactions.
- Classification by source: Natural (ascorbic acid, tocopherols, rosemary extract) and Synthetic (BHA, BHT, TBHQ).
- Classification by function: Primary antioxidants stop oxidation chain; Secondary ones chelate metals or decompose peroxides.
- Common natural antioxidants: Vitamin C (E300), Vitamin E (E306–309), carotenoids, and flavonoids.
- Synthetic antioxidants: Used in bakery, snacks, margarine, and frying oils for stability.
- Effectiveness depends on temperature, pH, oxygen, and concentration.
- Health aspects: Natural antioxidants are beneficial; synthetic types must follow ADI limits.

- Regulated by FSSAI/FAO/WHO Codex to ensure safe use levels.

19. Describe different types of microbial spoilage.

- Microbial spoilage is food deterioration caused by bacteria, yeasts, and molds.
- Major microbes: Bacteria (Bacillus, Clostridium), Yeasts (Saccharomyces, Candida), Molds (Aspergillus, Penicillium).
- Cereal foods: Mold growth and musty odor due to storage fungi.
- Fruits and vegetables: Soft rot, mold growth, and fermentation by yeasts.
- Meat and fish: Putrefaction, slime, and off-odors caused by proteolytic bacteria.
- Milk: Sour curdling and gas production by lactic acid bacteria.
- Yeast spoilage: Sugar fermentation leading to $CO_2$ and alcohol in juices and syrups.
- Mold spoilage: Surface growth producing toxins and color changes under aerobic conditions.
- Factors influencing spoilage: Temperature, pH, moisture, and oxygen.
- Prevention: Proper heat treatment, refrigeration, dehydration, and hygiene.

20. Explain the reasons behind food dehydration.

- Dehydration removes moisture from food to inhibit microbial and enzymatic activity.
- Purpose: Extends shelf life, reduces weight and volume for easy storage and transport.
- Principle: Reduction of water activity prevents microbial spoilage and enzyme reactions.
- Reasons: Prevent microbial growth, inactivate enzymes, and concentrate solids for convenience.
- Methods: Sun drying, hot air drying, spray drying, freeze drying, drum drying, vacuum drying.
- Quality effects: Proper dehydration retains nutrients, while overheating may cause color loss.
- Advantages: Lightweight, low-cost storage, long shelf life, and lower transportation cost.
- Factors affecting rate: Air temperature, humidity, airflow velocity, and food size.
- Applications: Used in dried fruits, vegetables, milk powder, instant coffee, and soups.
- Conclusion: Dehydration is a safe, economical preservation method ensuring longer shelf stability.

( 2x10 marks=20)